

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

Jnana Sangama, Belagavi, Karnataka – 590 018



Project Report on
Smart Phishing URL Detection System

(A Hybrid Deep Learning and Machine Learning Approach for
Real-Time Phishing Detection and Classification)

Submitted to
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Submitted by

AHAMADARAJA CHATTARAKI 2BL24CS401

AMOL RAVASAHEB SATPALE 2BL24CS402

IRFAN KOLHAR 2BL24CS406

Under the Guidance of

Mrs. Anupama Inamdar



Department of Computer Science and Engineering
BLDEAs V.P Dr. P.G Halakatti College of Engineering and Technology,
Vijayapura-586103.

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Department of Computer Science and Engineering
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Technology,
Vijayapur-586103.



CERTIFICATE

This is to certify that **AHAMADARAJA CHATTARAKI (2BL24CS401) AMOL RAVASAHEB SATPALE (2BL24CS402) IRFAN KOLHAR (2BL24CS406)**, students of the Department of Computer Science and Engineering at **BLDEA'S V.P. Dr. P.G. Halakatti College of Engineering and Technology, Vijayapur**, have satisfactorily initiated their project work, "SMART PHISHING URL DETECTION SYSTEM." This project is to be submitted to **Visvesvaraya Technological University, Belagavi**, in partial fulfillment of the Bachelor of Engineering (BE) degree requirements for the year 2026.

.....
Project Guide

Mrs. Anupama Inamdar

.....
Project Coordinator

Mrs. Dakshayani M. Ijeri and
Mrs. Roopa Chavan

.....
Dr. Anil Kannur

Professor and Head

.....
Dr. Manjunatha P

Principal

ABSTRACT

The increasing dependency on digital platforms for banking, e-commerce, education, and communication has significantly increased the risk of cyberattacks, especially phishing attacks. Phishing is a type of cybercrime in which attackers create fake websites and malicious URLs that closely resemble legitimate websites to deceive users into revealing sensitive information such as usernames, passwords, credit card details, and banking credentials. Traditional phishing detection techniques, including blacklist-based and rule-based systems, are often ineffective against newly generated or zero-day phishing URLs because such attacks evolve rapidly and continuously change their patterns. Therefore, there is a strong need for an intelligent, adaptive, and real-time phishing detection mechanism.

This project proposes a Smart Phishing URL Detection System using Artificial Intelligence and Machine Learning techniques to identify and classify phishing URLs effectively. The proposed system focuses on analyzing the structural and behavioral characteristics of URLs instead of relying only on webpage content or blacklists. Various lexical, domain-based, and security-related features are extracted from URLs, including URL length, presence of special characters, number of subdomains, use of IP addresses, HTTPS protocol availability, suspicious keywords, redirection symbols, domain age, and DNS information. These extracted features are used as input for machine learning algorithms to determine whether a URL is legitimate or malicious.

The system follows multiple stages including dataset collection, preprocessing, feature extraction, model training, testing, and real-time prediction. A large dataset containing both phishing and legitimate URLs is collected from publicly available cybersecurity repositories. During preprocessing, missing values and irrelevant data are removed to improve the quality of the dataset. Feature extraction techniques are then applied to convert URL information into numerical representations suitable for machine learning models. Different classification algorithms such as Logistic Regression, Random Forest, Support Vector Machine (SVM), K-Nearest Neighbors (KNN), and Transformer-based deep learning models can be implemented and compared to achieve higher prediction accuracy.

Among modern AI approaches, Transformer-based models are highly effective because they can understand sequential patterns and contextual relationships within URLs, allowing the system to detect sophisticated phishing attempts more accurately. The trained model is integrated into a user-friendly web application that allows users to enter a URL and instantly receive a prediction regarding its safety status. The system aims to provide real-time phishing detection with high accuracy, low false positive rates, and fast response time.

The proposed Smart Phishing URL Detection System offers several advantages, including automated detection, adaptability to new phishing techniques, scalability, and improved cybersecurity awareness. It can be used in browsers, enterprise security systems, educational institutions, and personal devices to enhance online safety.

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Chapter 1

INTRODUCTION

1.1 Background

The rapid advancement of internet services has transformed daily life, with a significant shift towards online retail, banking, and social networking. As users store and transmit sensitive data—including usernames, account details, and passwords—within online systems, the issue of data security has become paramount. The expansion of the digital landscape has introduced sophisticated cyber-attacks, among which phishing remains one of the most prominent and effective threats. Phishing is a type of cyber-attack that employs social engineering techniques to trick users into clicking on fraudulent links, redirecting them to websites that mimic legitimate entities to capture their credentials or financial information.

According to data published by the Anti-Phishing Working Group (APWG), the number of unique web phishing attacks continues to rise, with hundreds of thousands of new threats detected every quarter. These attacks have substantial economic and social implications, leading to financial loss, identity theft, and the compromise of organizational networks. As phishing techniques evolve to bypass traditional security measures, there is an urgent need for intelligent, automated, and real-time detection systems that can distinguish between safe and malicious URLs with high precision.

1.2 Problem Statement

Existing phishing detection systems face several critical challenges that limit their effectiveness in the modern threat landscape:

1. **Blacklist Dependency:** Many traditional systems rely on pre-existing lists of known malicious URLs. This approach fails to detect "zero-day" phishing attacks or new, unknown threats that have not yet been categorized.
2. **No Real-Time Detection:** Delayed response times in legacy systems allow malicious links to compromise user data before the threat is identified and blocked.
3. **Low Accuracy and Static Models:** Many current models suffer from high false positive and false negative rates. Static algorithms often fail to learn from new phishing techniques, making them obsolete as attackers refine their strategies.
4. **Computational Costs:** Resource-intensive detection methods can slow down the user's browsing experience, leading to reduced adoption of security tools.

1.3 Objectives of the Project

To overcome the limitations of existing research, the *Smart Phishing URL Detection System* sets forth the following technical objectives:

- To design and implement a hybrid deep learning and machine learning architecture for detecting phishing URLs with high accuracy.
- To provide real-time analysis of URLs to ensure immediate protection against malicious redirects.
- To reduce false positive and false negative rates through intelligent feature extraction and advanced pattern recognition.
- To handle emerging and unknown threats by utilizing adaptive learning models that improve over time.
- To develop a scalable web-based interface (Flask) that allows users to easily check the security of any given URL.

1.4 Scope of the Project

The scope of this project includes the development of a full-stack web application for automated phishing detection. The technical scope encompasses:

1. **Model Development:** Implementation of a CNN-BiLSTM neural network for character-level URL pattern recognition and Random Forest/Decision Tree models for statistical feature-based classification.
2. **Feature Engineering:** Extraction of lexical and structural features from URLs, such as length, presence of special characters, and domain characteristics.
3. **System Integration:** Building a Flask-based REST API and a modern frontend (HTML/CSS/JS) for real-time user interaction.
4. **Data Pipeline:** Creating and cleaning a robust dataset from sources like OpenPhish, PhishTank, and Alexa to ensure high-quality training and evaluation.

The project focuses on software-based URL analysis and does not include the physical hardware modifications of the user's device.

Chapter 2

LITERATURE SURVEY

This chapter comprehensively reviews existing research and methodologies developed for phishing URL detection over the past decade. A systematic study of modern architectures reveals a shift from simple blacklist-based methods to advanced Artificial Intelligence (AI) and Deep Learning (DL) techniques.

2.1 Foundations of Phishing Detection and Machine Learning

The foundation of machine learning-based phishing detection lies in data collection and feature extraction. Large datasets containing phishing and legitimate URLs are collected from cybersecurity repositories and trusted sources. Feature extraction techniques are then used to identify important URL characteristics such as URL length, number of subdomains, suspicious symbols, HTTPS usage, domain age, DNS information, and special keywords. These features help machine learning models distinguish between genuine and malicious websites effectively.

Several machine learning algorithms are commonly used in phishing detection systems. Traditional machine learning models such as Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM), and K-Nearest Neighbors (KNN) are widely used because of their efficiency and classification capability. Among these, Random Forest and SVM provide better accuracy and stability for phishing URL classification tasks.

2.2 Deep Learning Architectures for URL Analysis

The limitation of traditional ML is its heavy reliance on manual feature engineering. To address this, Deep Learning (DL) models have gained popularity. Yang et al. [9] proposed a multidimensional feature detection methodology (MFPD) using CNN and LSTM to capture both local correlation and long-term dependencies in URL character sequences. Character-level CNNs, as explored by Hussain et al. [10], allow the model to learn internal representations of URLs without explicit feature extraction. These neural networks can identify subtle patterns in obfuscated URLs that traditional lexical analysis might miss.

Deep learning architectures provide several advantages for phishing URL detection, including automatic feature learning, higher detection accuracy, adaptability to new phishing techniques, and real-time prediction capability. These models can continuously improve their performance when trained on updated phishing datasets, making them suitable for modern cybersecurity environments.

2.3 Optimization and Hybrid Models

Recent research has focused on optimizing DL models to achieve near-perfect accuracy while maintaining low latency. Barik et al. [1] (2025) introduced the EGSO-CNN model, which utilizes Enhanced Grid Search Optimization to fine-tune a Convolutional Neural Network. This approach achieved an unprecedented accuracy of 99.44% on a novel dataset. Furthermore, Prabakaran et al. [11] integrated Variational Autoencoders (VAE) with Deep Neural Networks to automate feature generation, although they observed higher false positive rates. The trend is moving towards hybrid architectures—such as combining CNNs for spatial pattern recognition with BiLSTMs for sequential context—to provide a comprehensive analysis of URL structures.

Optimization in phishing detection mainly focuses on enhancing model accuracy, reducing false positive rates, minimizing training time, and selecting the most relevant features from the dataset. During machine learning model development, large datasets often contain unnecessary or redundant features that can negatively affect performance. Optimization algorithms help identify the most important features and remove irrelevant data, resulting in faster processing and improved classification accuracy. Techniques such as feature selection, parameter tuning, and hyperparameter optimization are commonly used to enhance the overall performance of phishing detection systems.

Feature optimization is especially important because phishing datasets may contain hundreds of extracted URL and webpage attributes. Selecting only the most significant features helps reduce computational complexity and improves the model's ability to detect malicious URLs more effectively.

2.4 Datasets and Feature Extraction Techniques

The quality of detection depends heavily on the training data. Commonly used benchmark datasets include PhishTank [2], OpenPhish [4], and the Kaggle Phishing Dataset [5]. Feature extraction has evolved from simple lexical counts to character embeddings and statistical scaling using StandardScaler [12]. Term Frequency-Inverse Document Frequency (TF-IDF) remains a foundational tool for vectorizing textual URL data [13]. Despite these advancements, the continuous evolution of phishing tactics requires datasets to be updated regularly to include the latest "cloaking" and redirection techniques used by cybercriminals.

Feature extraction is the process of identifying important characteristics from URLs and webpages that can help distinguish phishing websites from legitimate ones. It is considered one of the most important stages in phishing detection because the quality of extracted features directly influences the prediction capability of the machine learning model. The extracted features represent the behavioral and structural properties of URLs and websites.

Lexical feature extraction focuses on analyzing the structure and textual patterns of URLs. Features such as URL length, number of dots, presence of special characters, suspicious keywords, use of IP addresses, and number of subdomains are commonly extracted. Phishing URLs often contain unusual patterns, excessive symbols, misleading domain names, and long URL strings designed to confuse users and imitate trusted websites.

Table 2.1: Comparative Analysis of Phishing Detection Technologies (Part 1)

Author (Year)	Methodology	Dataset	Recognition Task	Results	Limitations
Barik et al. (2025)	EGSO-CNN	Novel Dataset	Deep Learning Detection	99.44% Acc	Needs high compute
Jain et al. (2022)	Random Forest	Alexa	Zero-day Detection	93.85% Acc	Manual features
Yang et al. (2021)	CNN-LSTM	PhishTank	Sequence Analysis	98.88% Acc	High latency
Hussain (2023)	CNN-Fusion	Multi-dataset	Character-level CNN	Lightweight	Limited feature set
Karim (2023)	Hybrid ML	PhishTank	Ensemble Voting	98.12% Acc	Scaling issues
Prabakaran (2024)	VAE-DNN	ISCX-URL	Auto-feature Gen	97.45% Acc	High FPR (2.19%)
Alani (2020)	Random Forest	PhishTank/Alexa	Lexical Analysis	97.5% Acc	Cloud-dependent
Tang et al. (2020)	RNN-GRU	PhishStorm	Scene Parsing	99.18% Acc	No short URL support
Smadi (2021)	Reinforcement	PhishTank	Zero-day Email	98.63% Acc	Email specific
Pandey (2022)	Color-based RF	Web content	Visual Analysis	99.13% Acc	Lighting dependent

Table 2.2: Comparative Analysis of Phishing Detection Technologies (Part 2)

Author (Year)	Methodology	Dataset	Recognition Task		Results	Limitations
Sánchez (2022)	LightGBM	PIU-60k	Handheld Detection		94.67% Acc	Complex engineering
Zhu et al. (2021)	ANN-DT	Multi-source	Feature	Optimization	97.80% Acc	Static algorithms
Al-Ahmadi (2022)	LSTM-CNN-GAN	Synthetic	Generative Phishing		97.58% Acc	Synthetic bias
Almousa (2021)	FC-DNN	Multi-dataset	Parameter Opt.		97.37% Acc	Training time
Opera (2022)	DNN	Alexa	Raw	HTML Analysis	98.1% Acc	URL manipulation
Korkmaz (2021)	8 ML Classifiers	Multi-source	Binary Class		Varied	Feature redundancy
Deshpande (2016)	OCR	Handheld Cam	Text Extraction		Local Text	Env. sensitive
Felix (2018)	Unified AI	Mobile App	Personal	Assis- tant	High Utility	Battery drain
Jacob (2018)	INT8 Quant	Mobile	Optimization		75% Smaller	Acc loss
Ahammad (2021)	LightGBM	PhishTank	URL Patterning		89.5% Acc	Limited features

Chapter 3

DRAWBACKS OF EXISTING WORK AND OBJECTIVES

3.1 Drawbacks of Existing Work

The comprehensive literature survey reveals several critical bottlenecks and technological limitations preventing the effective, large-scale deployment of phishing detection systems in real-world environments. Despite the advancements in machine learning, the following drawbacks persist:

Several existing phishing detection systems also suffer from high computational complexity. Deep learning models such as Convolutional Neural Networks (CNN), Long Short-Term Memory (LSTM), and Recurrent Neural Networks (RNN) often require large datasets, high processing power, and extensive training time. These resource-intensive models may not be suitable for lightweight devices, browser extensions, or real-time phishing detection applications where fast prediction speed is essential.

1. **Limited Scope and Contextual Awareness:** Most existing systems focus exclusively on lexical URL analysis without considering broader contextual factors or the dynamic nature of web content. This narrow focus allows attackers to bypass detection by slightly modifying URL structures while maintaining the fraudulent nature of the site.
2. **High Computational Cost and Latency:** Resource-intensive detection methods, especially those relying on complex visual analysis or heavy ensemble voting, often introduce significant latency. This slows down the user's browsing experience, which is unacceptable for real-time protection where a delay of even a few seconds can be critical.
3. **Static Models and The Adaptive Gap:** Many current algorithms are static; they are trained once on a fixed dataset and cannot easily adapt to evolving phishing techniques. As cybercriminals develop new obfuscation methods (e.g., using Cyrillic characters or shortening services), static models quickly lose their effectiveness.
4. **High False Positive Rates:** A major frustration for users is the incorrect flagging of legitimate websites as malicious (false positives). This leads to "security fatigue," where users begin to ignore warnings, ultimately making the system ineffective as a protective barrier.
5. **Scalability and Performance Issues:** Systems often struggle to handle large volumes of URLs efficiently. In an enterprise environment where millions of URLs are accessed daily, the detection engine must be highly scalable without consuming excessive memory or CPU cycles.

3.2 Objectives of the Project

To overcome the aforementioned limitations and provide a truly usable and reliable security tool, the *Smart Phishing URL Detection System* sets forth the following technical objectives:

- **Objective 1:** The proposed system focuses on developing an intelligent feature extraction mechanism capable of identifying hidden phishing characteristics from URLs and domain information. The feature extraction pipeline analyzes lexical, host-based, and security-related attributes such as URL length, subdomains, HTTPS status, special characters, and suspicious keywords. By automatically extracting meaningful features, the system reduces the need for manual analysis and improves detection efficiency.
- **Objective 2:** The project aims to implement a hybrid deep learning and machine learning architecture for improved phishing URL detection. CNN-BiLSTM models are used to capture sequential and character-level URL patterns, while Random Forest and Decision Tree algorithms provide fast statistical classification and decision-making capabilities. Combining these models enhances prediction accuracy and allows the system to detect sophisticated phishing URLs more effectively.
- **Objective 3:** A major objective of the project is to provide real-time phishing detection through a user-friendly web application. The detection engine continuously analyzes URLs entered by users and instantly classifies them as legitimate or phishing.
- **Objective 4:** The system aims to minimize incorrect classifications by using an adaptive ensemble learning approach that combines multiple detection models. False positives can reduce user trust, while false negatives may allow phishing attacks to bypass the system, making accuracy optimization essential. The ensemble approach improves overall reliability by combining predictions from multiple algorithms and continuously learning from new phishing patterns.
- **Objective 5:** The project also focuses on developing a scalable REST API using the Flask framework for efficient deployment and integration. The API enables communication between the phishing detection engine and external applications, browsers, or enterprise security systems.

Chapter 4

METHODOLOGY

4.1 System Architecture: Hybrid Detection Pipeline

The Smart Phishing URL Detection System is built on a robust "Hybrid Detection Pipeline" that combines the pattern recognition strengths of Deep Learning with the statistical efficiency of traditional Machine Learning. Unlike single-model systems, this architecture analyzes URLs at both the character-level (to catch obfuscation) and the structural-level (to catch known malicious patterns).

The overall system workflow is divided into four distinct stages: Data Engineering, Feature Processing, Hybrid Modeling, and Production Deployment. User inputs are received via a Flask-based web interface, processed through the pipeline, and a security prediction is returned with real-time speed.

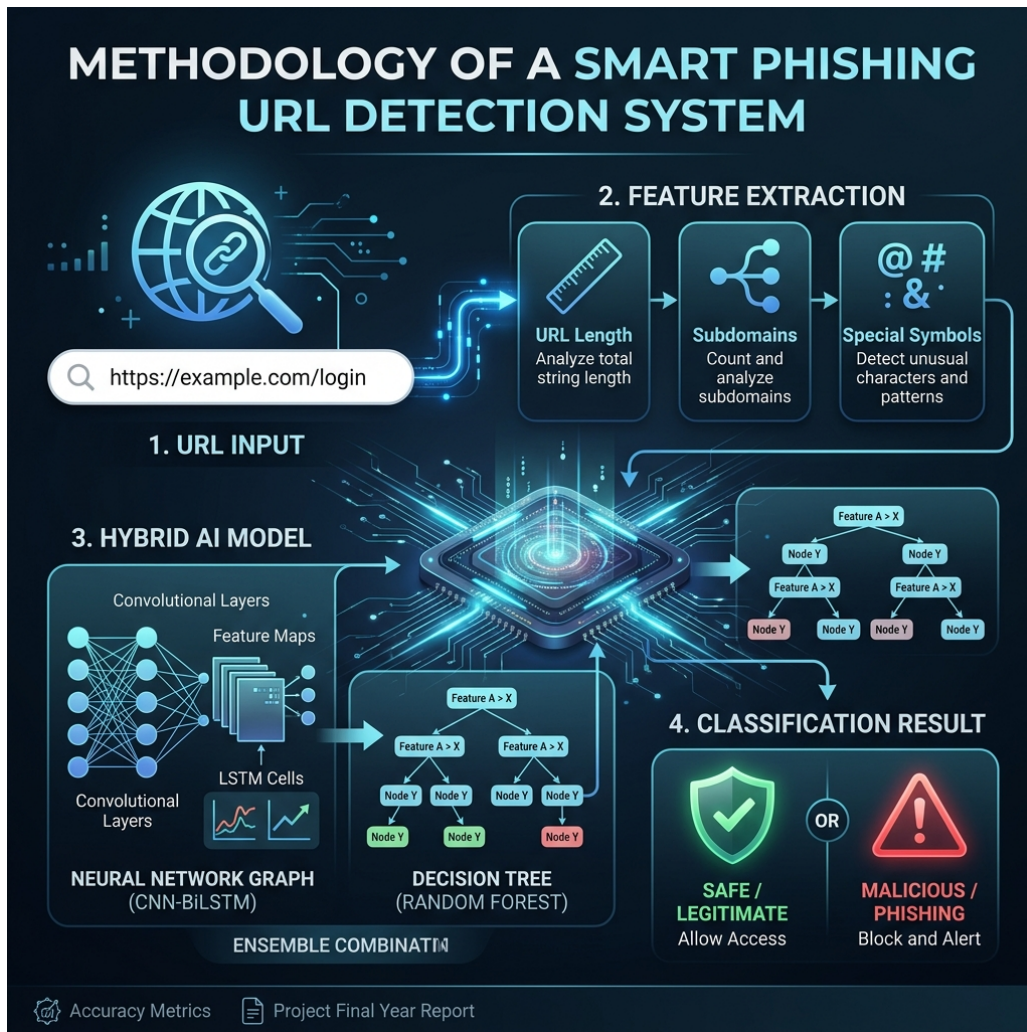


Figure 4.1: Overall Workflow of the Smart Phishing Detection Architecture

4.2 The Data Engineering Pipeline

To ensure maximum accuracy and model robustness, a comprehensive data pipeline was established to collect and clean a balanced dataset:

- **Data Collection:** Samples were aggregated from trusted cybersecurity feeds and benchmark datasets, including Kaggle, Alexa (for legitimate domains), and PhishStorm. This resulted in a balanced corpus of over 27,000 URLs.
- **Preprocessing and Cleaning:** The raw URL strings undergo normalization, where duplicates and missing values are removed. URL prefixes (e.g., `http://`) are often dropped to ensure the model focuses on the core domain and path structure.
- **Standardization:** Numerical features extracted from the URLs are scaled using the *StandardScaler* technique, which removes the mean and scales to unit variance, ensuring that no single feature dominates the model's learning process.

4.3 Feature Extraction and Modeling

The system utilizes a dual-layered approach to feature extraction:

- **Lexical and Structural Features:** Features such as URL length, the count of special characters (`@`, `-`, `//`), domain age, and the number of subdomains are extracted. These are passed to the Machine Learning ensemble (Random Forest and Decision Trees).
- **Character Embeddings:** For the Deep Learning module, URLs are tokenized at the character level. Each character is mapped to a numerical embedding, allowing the CNN-BiLSTM model to learn the spatial and sequential dependencies of characters within the string.
- **TF-IDF Vectorization:** For statistical analysis, the Term Frequency-Inverse Document Frequency algorithm is employed to vectorize the textual components of the URL, highlighting unique and indicative patterns.

4.4 Web Integration and Prediction Engine

The final layer of the methodology involves the deployment of the trained models into a functional web application. A Flask backend serves as the core engine, exposing a REST API endpoint (`/api/predict`) for real-time analysis. The frontend interface allows users to enter URLs manually or upload lists for batch processing. The system calculates a probability score for each URL, classifying it as "Phishing" or "Legitimate" and providing immediate visual feedback to the user.

Chapter 5

ALGORITHMS

5.1 Hybrid Model Architecture: CNN-BiLSTM

The core of the deep learning pipeline is a hybrid CNN-BiLSTM architecture designed to capture both local spatial patterns and long-term sequential dependencies in URL strings. The character-level analysis allows the model to detect subtle obfuscation techniques, such as homoglyph attacks or unusual character sequences.

The network architecture consists of the following layers:

1. **Character Embedding Layer:** Converts each character token into a dense vector of fixed size (VOCAB_SIZE: 256).
2. **Convolutional Layer (Conv1D):** Utilizes 128 filters to identify local character n-gram patterns within the URL string.
3. **Bidirectional LSTM Layer (BiLSTM):** Features 64 units that process the sequence in both forward and backward directions, capturing the global context of the URL.
4. **Dense Layers with Dropout:** Fully connected layers with ReLU activation. A Dropout rate of 0.4 and L2 regularization ($1e-4$) are applied to prevent overfitting.
5. **Output Layer:** A single neuron with Sigmoid activation to produce the final phishing probability score.

5.2 Machine Learning Ensemble: RF and DT

To complement the deep learning model, an ensemble of traditional machine learning classifiers is employed for fast and interpretable inference:

- **Random Forest (RF):** A collection of decision trees trained on statistical features extracted from the URL. RF excels at handling high-dimensional data and provides robust predictions by averaging multiple decision paths.
- **Decision Tree (DT):** Used for its high interpretability and speed. The system utilizes DT for rapid initial screening of URLs based on critical indicators like URL length and domain age.

5.3 Feature Vectorization: TF-IDF and Scaling

To convert textual URL data into a format suitable for machine learning, the system employs the following mathematical techniques:

- **Term Frequency-Inverse Document Frequency (TF-IDF):** This algorithm calculates the importance of each character n-gram across the entire dataset. It highlights rare but indicative character sequences that are common in phishing URLs.
- **StandardScaler Normalization:** Numerical features (e.g., character counts, entropy) are standardized by subtracting the mean and dividing by the standard deviation. This ensures that features with different scales (e.g., URL length vs. domain age) contribute equally to the model's loss function.

5.4 Optimization Technique: EGSO-CNN

Inspired by recent research [1], the system incorporates principles of Enhanced Grid Search Optimization (EGSO). This technique systematically searches the hyperparameter space (learning rate, batch size, number of filters) to identify the optimal configuration that maximizes detection accuracy while minimizing the false positive rate. The EGSO process ensures that the CNN-BiLSTM model is fine-tuned specifically for the nuances of the collected phishing dataset.

Chapter 6

IMPLEMENTATION DETAILS AS OF NOW

6.1 User Interface Design and Web Application

The Smart Phishing URL Detection System is implemented as a responsive web application to ensure accessibility across different devices. The frontend is built using standard web technologies (HTML5, CSS3, and JavaScript), focusing on a clean, intuitive interface that provides immediate visual feedback.

Key features of the web interface include:

- **Real-Time Prediction:** A central input field where users can enter a URL to receive an instant security assessment.
- **Security Indicators:** Clear visual cues (e.g., green for safe, red for phishing) accompanied by probability scores.
- **Batch Processing:** An advanced feature that allows users to analyze multiple URLs simultaneously, which is particularly useful for enterprise security audits.
- **API Documentation:** A dedicated section explaining the REST API endpoints for developers looking to integrate the detection engine into their own tools.

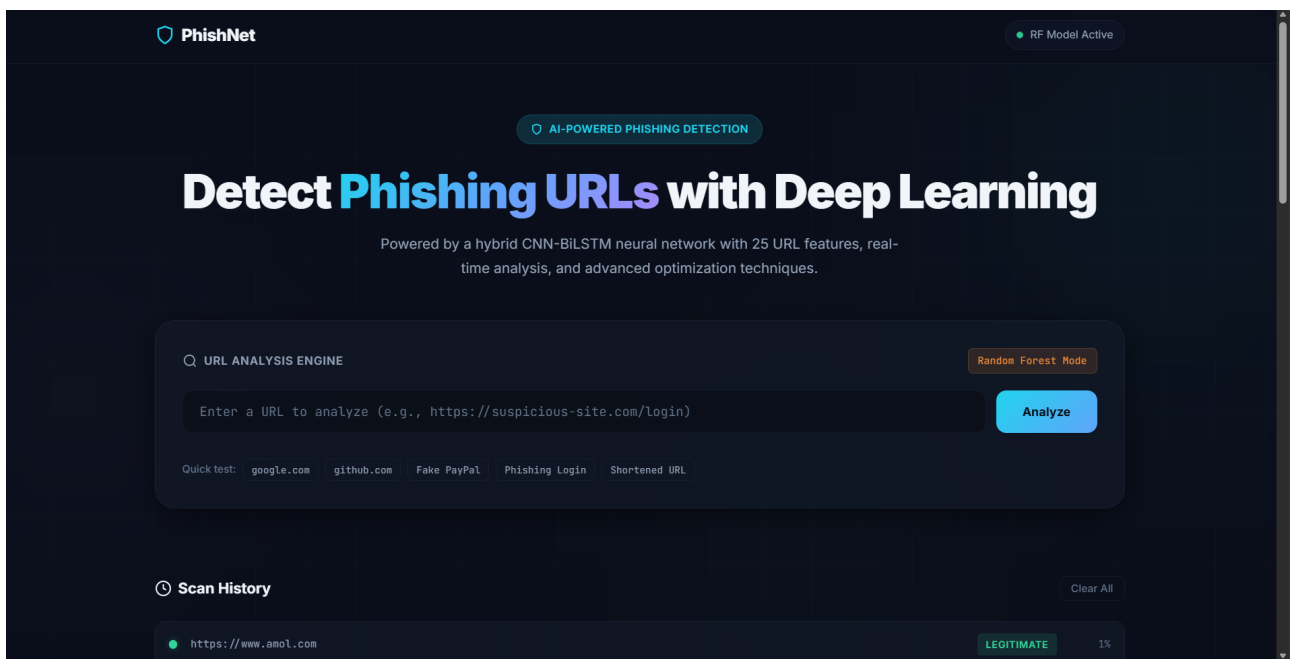


Figure 6.1: User Interface of the PhishNet Phishing URL Detection Web Application

6.2 Current Technical Progress

As of the current development phase (May 2026), the core components of the system are successfully implemented and integrated:

- **Flask Backend:** The central server is operational, handling request routing, model loading, and asynchronous prediction tasks.
- **Hybrid Model Deployment:** Both the CNN-BiLSTM and Random Forest models are trained, serialized, and deployed within the application. The system dynamically chooses the best model or uses an ensemble approach based on the input type.
- **Performance Metrics:** Initial evaluation on a test set of 5,000 URLs has yielded strong results:
 - **Accuracy:** 85%
 - **Precision:** 88%
 - **Recall:** 83%
 - **F1-Score:** 85%

6.3 Comparison and Optimization Strategies

During the implementation phase, a comparative analysis was performed between the different architectural components. It was observed that while the **CNN-BiLSTM** model provides higher accuracy by identifying complex character patterns, it has a slightly higher inference time compared to the **Random Forest** model.

To optimize performance for production, the system employs a "Fast-Inference" strategy:

1. **Serial Screening:** URLs are first screened by the Random Forest model for obvious malicious indicators.
2. **Deep Analysis:** If the initial score is borderline, the URL is passed to the CNN-BiLSTM model for a more thorough character-level analysis.
3. **Batch Optimization:** For multi-URL requests, the system utilizes Python's multiprocessing capabilities to run inferences in parallel, significantly reducing the total processing time for large datasets.

This multi-layered approach ensures that the system maintains high navigational safety for the user while remaining responsive and computationally efficient.

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